

AIMS System - Detailed Class Design Documentation

12. CartItem Class Design

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	productId	int	null	ID of the product in the cart
2	quantity	int	0	Quantity of the product
3	cartID	int	null	Unique identifier for the cart
4	cartItemID	int	null	Unique identifier for the cart item

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	changeQuantityOfProducts.	void	Updates the quantity and recalculates subtotal
2	addProductToCart	void	Adds a product to the cart with specified quantity

Parameter:

- addProductToCart(product: Product, quantity: int): voids
 - **product:** A Product object representing the product to add.
 - **quantity:** An integer representing the number of products to add.
 - **Return:** None.
- changeQuantityofProduct(product: Product, quantity: int): void
 - **product:** A Product object representing the product to update.
 - **quantity:** An integer representing the new quantity.
 - **Return:** None.

Exception:

- **addProductToCart(productId: int, quantity: int): void**
 - **ProductNotFoundException:** Raised if the specified product does not exist in the system.
 - **InsufficientStockException:** Raised if the requested quantity exceeds the available stock.
- **changeQuantityOfProduct(productId: int, quantity: int): void**
 - **ProductNotFoundException:** Raised if the product is not found in the cart.
 - **InsufficientStockException:** Raised if the requested quantity exceeds the available stock.

Method:

How to use parameters/attributes:

- **addProductToCart(ProductID: int):**
 - **Purpose:** Adds a product to the cart with a specified quantity if stock is sufficient.
 - **Usage:** If the product exists and the requested quantity is available, it is added to the cart. Otherwise, InsufficientStockException is raised.
- **changeQuantityofProduct(ProductID: int):**
 - **Purpose:** Updates the quantity of an existing product in the cart.
 - **Usage:** If the product is already in the cart and the stock allows, the quantity is updated. Otherwise, InsufficientStockException is raised.

State

- **Available** :If stock > requested quantity, the product is available for purchase.
- **Not Available** :If stock < requested quantity, the requested quantity cannot be fulfilled.
- **Product Not Found**:If product_id is invalid (i.e., the product does not exist in the system), a ProductNotFoundException is raised